University of Hamburg

Faculty of Business Administration
Institute of Logistics, Transport and Production

Seminar Paper

Title of the Thesis

Independent academic work
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List of Abbreviations

LLM : Large Language Model

OR : Operations Research

VRP : Vehicle Routing Problem

List of Symbols

$\mathcal{I}$: Set of customers

$\mathcal{J}$: Set of all locations (customers and depot)

K : Number of vehicles

c_{ij} : Travel cost from location i to location j

d_i : Demand of customer $i \in \mathcal{I}$

Q : Vehicle capacity

x_{ij} : Binary variable, 1 if a vehicle travels from i to j

1 Introduction

Operations research provides powerful tools for decision-making in complex systems. In recent years, advances in computational methods have enabled the solution of increasingly large-scale optimization problems (Desaulniers et al., 2005). This paper addresses the problem of … .

Figure 1 shows a schematic overview of the problem structure.

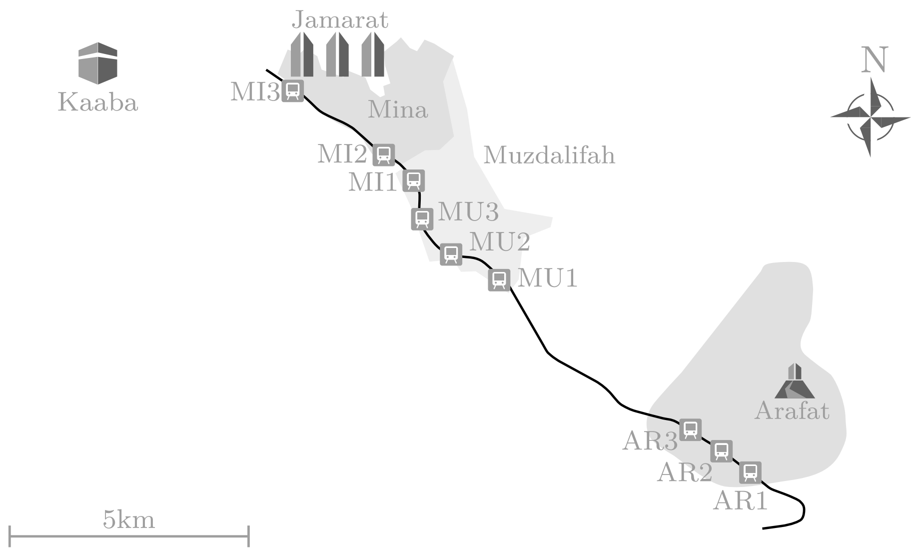

Figure 1: Replace this with your own figure

The remainder of this paper is structured as follows. Section 2 reviews the relevant literature. Section 3 introduces the mathematical model, including the underlying assumptions in Section 3.1 and the formal formulation in Section 3.2. Section 4 presents the computational study, and Section 6 concludes with a summary and outlook for future research.

2 Literature Review

The vehicle routing problem has been studied extensively since the seminal work of Dantzig & Ramser (1959). Several exact and heuristic approaches have been proposed over the decades (Toth & Vigo, 2014).

Column generation has proven to be a particularly effective technique for solving large-scale routing and scheduling problems (Desaulniers et al., 2005). Recent advances in machine learning have also been applied to improve heuristic performance (Bengio et al., 2021).

A comprehensive overview of solution methods is beyond the scope of this paper. For a detailed survey, the reader is referred to Toth & Vigo (2014).

3 Model

This section introduces the mathematical model for the problem described above. We first state the key assumptions, then present the formal formulation.

3.1 Assumptions

The modeling of the problem requires several simplifying assumptions:

- A single depot serves as the starting and ending point for all vehicles.
- Each customer $i \in \mathcal{I}$ has a known demand d_i that must be fully satisfied.
- The fleet consists of K identical vehicles, each with capacity Q.
- Travel costs c_{ij} between locations i and j are symmetric and satisfy the triangle inequality.

3.2 Formulation

We define the following notation:

Table 1: Notation overview

Symbol	Description
$\mathcal{I}$	Set of customers
$\mathcal{J}$	Set of all locations (customers and depot)
K	Number of vehicles
c_{ij}	Travel cost from location i to location j
d_i	Demand of customer $i \in \mathcal{I}$
Q	Vehicle capacity
x_{ij}	Binary variable, 1 if a vehicle travels from i to j

The model is defined as follows:

$$\min \sum_{i \in \mathcal{J}} \sum_{j \in \mathcal{J}} c_{ij} \cdot x_{ij} \tag{1}$$

subject to:

$$\sum_{j \in \mathcal{J}} x_{ij} = 1 \qquad \forall i \in \mathcal{I} \tag{2}$$

$$\sum_{i \in \mathcal{J}} x_{ij} = 1 \qquad \forall j \in \mathcal{I} \tag{3}$$

$$x_{ij} \in \{0, 1\} \qquad \forall i, j \in \mathcal{J} \tag{4}$$

The objective function (Equation 1) minimizes total travel cost. Equation 2 ensures that each customer is visited exactly once, while Equation 3 guarantees flow conservation. Equation 4 defines the binary decision variables. The notation is summarized in Table 1.

4 Computational Study

To evaluate the performance of the proposed approach, we conduct a series of computational experiments. All experiments were performed on a machine with an Intel Core i7 processor and 16 GB of RAM. The model was implemented in Python using Gurobi 11.0 as the solver.

Table 2: Computational results comparing exact and heuristic solutions

Instance	$\mid \mathcal{I} \mid$	Optimal	Heuristic	Gap (%)	Time (s)
Small-1	25	245.3	248.1	1.14	0.8
Small-2	25	312.7	318.4	1.82	1.1
Medium-1	50	1,340.5	1,382.9	3.16	12.4
Medium-2	50	1,128.3	1,167.2	3.44	15.7
Large-1	100	3,450.1	3,621.3	4.96	245.3
Large-2	100	4,217.8	4,485.6	6.35	310.2

The results in Table 2 show that the heuristic approach finds near-optimal solutions within reasonable computation times. For small instances, the optimality gap remains below 2%, while for larger instances the gap increases to approximately 5–6%.

5 Discussion

The results indicate that the proposed heuristic performs well for small and medium instances but shows increasing gaps for larger problem sizes. This is consistent with findings by Toth & Vigo (2014), who observed similar scaling behavior for constructive heuristics applied to capacitated routing problems.

A key limitation of this study is the use of symmetric cost matrices, which may not hold in real-world logistics networks with one-way streets or time-dependent travel times. Furthermore, the test instances were generated randomly rather than derived from real-world data, which limits the generalizability of the results.

6 Conclusion and Outlook

This paper presented a mathematical model and heuristic solution approach for the vehicle routing problem. The computational study demonstrated that the proposed method achieves near-optimal solutions with significantly reduced computation times compared to exact methods.

Several directions for future research emerge from this work. First, the model could be extended to incorporate time windows and heterogeneous fleets. Second, the heuristic could be enhanced with machine learning techniques to improve initial solution quality. Finally, real-world case studies would help validate the practical applicability of the approach.

Declaration of Authorship

I hereby declare that I have written this thesis independently and without outside help, and that I have not used any sources or aids other than those indicated in the attached bibliography. All passages taken verbatim or in substance from published or unpublished sources are identified as such. All internet sources are attached to the thesis. Furthermore, I declare that I have not previously submitted this thesis in any other examination procedure and that the submitted written version corresponds to the version on the electronic storage medium.

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Use of AI Tools

The following AI tools were used during the preparation of this thesis:

- **Tool 1**: Used for … .
- **Tool 2**: Used for … .
- **Tool 3**: Used for … .

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